

Mathematical Methods I

Differential operators

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1. Musical isomorphisms

On a manifold with a nondegenerate metric g , the **musical isomorphisms** identify tangent vectors with covectors. The flat sign $\flat$ lowers an index, and the sharp sign $\sharp$ raises it. The names come from musical notation.

In Cartesian orthonormal coordinates, one may mistake this identification for transposing a column into a row. A direct transpose has no invariant meaning in general curvilinear coordinates: the metric supplies the conversion and respects coordinate changes.

Flat: vectors to 1-forms

For $X \in T_p M$, define the covector $X^\flat \in T_p^* M$ by its action on any $v \in T_p M$:

$$X^\flat(v) = g(X, v).$$

If $X = X^j \partial_j$ in coordinates u^i , then

$$X^\flat = g_{ij} X^j du^i, \quad (X^\flat)_i = g_{ij} X^j.$$

The matrix representing $\flat$ in these bases is (g_{ij}) .

Sharp: 1-forms to vectors

Nondegeneracy makes (g_{ij}) invertible; its inverse has entries g^{ij} . For $\alpha = \alpha_j du^j$, the inverse map $\sharp = \flat^{-1}$ gives

$$\alpha^\sharp = g^{ij} \alpha_j \partial_i, \quad (\alpha^\sharp)^i = g^{ij} \alpha_j.$$

Consequently, $(X^\flat)^\sharp = X$ and $(\alpha^\sharp)^\flat = \alpha$.

2. The gradient and the differential

The differential df of a scalar function is naturally a 1-form. The **gradient** is the unique vector field representing that 1-form through the metric:

Metric definition of the gradient

$$df(v) = g(\text{grad } f, v) \quad \text{for every tangent vector } v.$$

Equivalently,

$$\text{grad } f = (df)^\sharp = g^{ij} \frac{\partial f}{\partial u^j} \partial_i, \quad (\text{grad } f)^\flat = df.$$

In Cartesian orthonormal coordinates, $g_{ij} = \delta_{ij}$, so the components of df and $\text{grad } f$ happen to be numerically equal. In curvilinear coordinates, the inverse metric g^{ij} is essential.

Derivation in a coordinate basis

Write the unknown gradient as $\text{grad } f = A^i \partial_i$. Evaluating its defining equation on the basis vector ∂_j gives

$$\begin{aligned} g(\text{grad } f, \partial_j) &= A^i g(\partial_i, \partial_j) = A^i g_{ij}, \\ df(\partial_j) &= \partial_j f. \end{aligned}$$

Thus $A^i g_{ij} = \partial_j f$. Multiplying by g^{jk} yields $A^k = g^{kj} \partial_j f$, proving the coordinate formula. For a Euclidean region parametrized by u^i , the metric itself follows from the coordinate basis:

$$g_{ij} = \partial_i \mathbf{x} \cdot \partial_j \mathbf{x} = \sum_k \frac{\partial x^k}{\partial u^i} \frac{\partial x^k}{\partial u^j}.$$

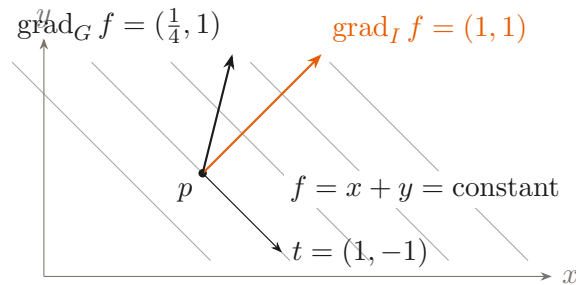


Figure 1: The same differential, two metrics. For $f = x + y$, $df = dx + dy$ is unchanged. With $I = \text{diag}(1, 1)$ the gradient is $(1, 1)$; with $G = \text{diag}(4, 1)$ it is $(1/4, 1)$. Both satisfy $g(\text{grad } f, t) = df(t) = 0$ along a level line. Orthogonality is measured by the chosen metric, not by the appearance on the page. Both arrows use the same display scale.

3. Orthogonal coordinates and scale factors

For orthogonal coordinates, $g_{ij} = 0$ whenever $i \neq j$. Define $h_i = \sqrt{g_{ii}} = \|\partial_i\|$, without summing over i . Then $g^{ii} = 1/h_i^2$. The same gradient has two useful component expressions:

Coordinate and orthonormal bases

In the natural coordinate basis $\{\partial_i\}$,

$$\text{grad } f = \sum_i \frac{1}{h_i^2} \frac{\partial f}{\partial u^i} \partial_i.$$

In the orthonormal basis $\hat{e}_i = \partial_i/h_i$,

$$\text{grad } f = \sum_i \frac{1}{h_i} \frac{\partial f}{\partial u^i} \hat{e}_i.$$

The coefficients differ because ∂_i has length h_i , while $\hat{e}_i$ has unit length.

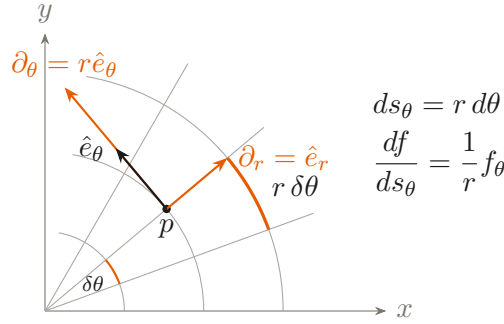


Figure 2: Why an angular derivative needs a scale factor. Equal angular increments span longer arcs at larger radii. At the illustrated point $r = 2$, the coordinate vector ∂_θ has length 2, while $\hat{e}_\theta$ has length 1. Thus the physical angular gradient component is f_θ/r ; its coordinate-basis coefficient is f_θ/r^2 .

4. Gradient in familiar coordinates

The formulas below distinguish the natural coordinate basis from the orthonormal basis used for physical vector components.

Polar coordinates (r, θ)

For $x = r \cos \theta$, $y = r \sin \theta$, the line element is $ds^2 = dr^2 + r^2 d\theta^2$, so $h_r = 1$ and $h_\theta = r$:

$$\begin{aligned} \text{grad } f &= \frac{\partial f}{\partial r} \partial_r + \frac{1}{r^2} \frac{\partial f}{\partial \theta} \partial_\theta, \\ &= \frac{\partial f}{\partial r} \hat{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{e}_\theta. \end{aligned}$$

Cylindrical coordinates (ρ, ϕ, z)

Here $x = \rho \cos \phi$, $y = \rho \sin \phi$, and $h_\rho = 1$, $h_\phi = \rho$, $h_z = 1$. Hence

$$\begin{aligned}\text{grad } f &= f_\rho \partial_\rho + \frac{f_\phi}{\rho^2} \partial_\phi + f_z \partial_z, \\ &= f_\rho \hat{e}_\rho + \frac{f_\phi}{\rho} \hat{e}_\phi + f_z \hat{e}_z.\end{aligned}$$

We write $f_\rho = \partial f / \partial \rho$, and similarly for the other partial derivatives.

Spherical coordinates (r, θ, ϕ)

With $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$,

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2, \quad (h_r, h_\theta, h_\phi) = (1, r, r \sin \theta).$$

Therefore,

$$\begin{aligned}\text{grad } f &= f_r \partial_r + \frac{f_\theta}{r^2} \partial_\theta + \frac{f_\phi}{r^2 \sin^2 \theta} \partial_\phi, \\ &= f_r \hat{e}_r + \frac{f_\theta}{r} \hat{e}_\theta + \frac{f_\phi}{r \sin \theta} \hat{e}_\phi.\end{aligned}$$

5. The Hodge star

On an oriented n -dimensional manifold with a nondegenerate metric, the **Hodge star** sends a p -form to an $(n - p)$ -form. Both spaces have dimension $\binom{n}{p} = \binom{n}{n-p}$.

Defining relation

For p -forms α and β , the metric induces an inner product on forms, and the star is characterized by

$$\alpha \wedge \star \beta = \langle \alpha, \beta \rangle_g \text{vol}_g.$$

Locally, with the chosen orientation,

$$\text{vol}_g = \sqrt{|\det(g_{ij})|} du^1 \wedge \cdots \wedge du^n, \quad \star 1 = \text{vol}_g.$$

The metric and orientation together determine $\star$.

Cartesian basis in Euclidean $\mathbb{R}^3$

For the standard orientation $dx \wedge dy \wedge dz$,

$$\begin{aligned} \star 1 &= dx \wedge dy \wedge dz, \\ \star dx &= dy \wedge dz, & \star dy &= dz \wedge dx, & \star dz &= dx \wedge dy, \\ \star(dy \wedge dz) &= dx, & \star(dz \wedge dx) &= dy, & \star(dx \wedge dy) &= dz, \\ \star(dx \wedge dy \wedge dz) &= 1. \end{aligned}$$

Orthonormal coframes in curvilinear coordinates

In orthogonal coordinates, the 1-forms $\vartheta^i = h_i du^i$ form an orthonormal coframe. The star acts on these forms as it does on dx, dy, dz in Cartesian coordinates; the scale factors appear when we return to du^i .

Polar plane

For $ds^2 = dr^2 + r^2 d\theta^2$ and orientation $dr \wedge d\theta$,

$$\text{vol}_g = r dr \wedge d\theta, \quad \star dr = r d\theta, \quad \star(r d\theta) = -dr, \quad \star(dr \wedge d\theta) = \frac{1}{r}.$$

Cylindrical space

With $\vartheta^1 = d\rho$, $\vartheta^2 = \rho d\phi$, $\vartheta^3 = dz$ and orientation $d\rho \wedge d\phi \wedge dz$,

$$\begin{aligned} \text{vol}_g &= \rho d\rho \wedge d\phi \wedge dz, \\ \star d\rho &= \rho d\phi \wedge dz, \\ \star(\rho d\phi) &= dz \wedge d\rho, \\ \star dz &= d\rho \wedge (\rho d\phi). \end{aligned}$$

Spherical space

With $\vartheta^1 = dr$, $\vartheta^2 = r d\theta$, $\vartheta^3 = r \sin \theta d\phi$,

$$\begin{aligned} \text{vol}_g &= r^2 \sin \theta dr \wedge d\theta \wedge d\phi, \\ \star dr &= r^2 \sin \theta d\theta \wedge d\phi, \\ \star(r d\theta) &= r \sin \theta d\phi \wedge dr, \\ \star(r \sin \theta d\phi) &= r dr \wedge d\theta. \end{aligned}$$

Applying the star twice

On a p -form in dimension n , if the metric has q negative directions,

Double-star identity

$$\star \star \alpha = (-1)^{p(n-p)+q} \alpha.$$

For a Riemannian metric, $q = 0$. Thus $\star^2 = 1$ on every degree in Euclidean $\mathbb{R}^3$. With Lorentzian signature $(-, +, +, +)$, $q = 1$, so $\star^2 = -1$ on 2-forms in four dimensions.

6. Divergence from forms and volume

The **divergence** of a vector field measures its local net outward flux per unit volume. The familiar Cartesian expression $\partial_x F_x + \partial_y F_y + \partial_z F_z$ relies on a constant orthonormal basis. Curvilinear coordinates also change the coordinate volume element.

On an oriented Riemannian manifold, the metric and the volume form provide a coordinate-free construction. First lower the vector index, then use the Hodge star to turn the 1-form into a flux $(n-1)$ -form:

$$V \xrightarrow{\flat} V^\flat \xrightarrow{\star} \star V^\flat \xrightarrow{d} d(\star V^\flat).$$

The final n -form is a scalar multiple of the volume form. Its coefficient is the divergence:

Intrinsic and coordinate formulas

$$d(\star V^\flat) = (\operatorname{div} V) \operatorname{vol}_g, \quad \operatorname{div} V = \star d \star V^\flat.$$

If $V = V^i \partial_i$ and $g = \det(g_{ij}) > 0$, then

$$\operatorname{div} V = \frac{1}{\sqrt{g}} \partial_i (\sqrt{g} V^i).$$

Here the second star extracts a scalar from the top-degree form. The formula also follows from $\mathcal{L}_V \operatorname{vol}_g = (\operatorname{div} V) \operatorname{vol}_g$.

The construction explains the role of each operation: $\flat$ gives the covariant 1-form, $\star$ turns it into a flux form, d records the outward flux, and the final $\star$ returns a scalar.

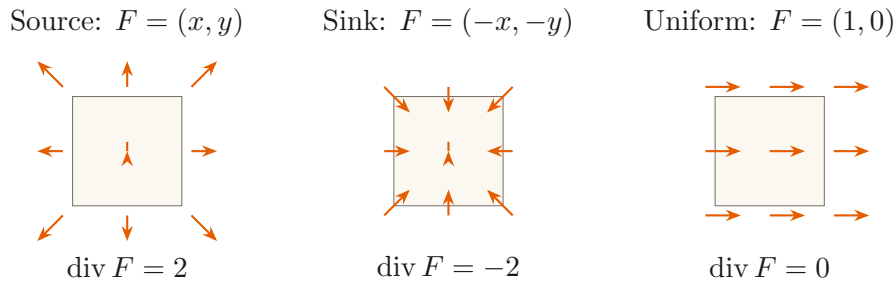


Figure 3: Divergence counts net flux, not field strength. The squares are control regions: a source has excess outflow, a sink excess inflow, and a uniform field has balanced flux. In the Euclidean plane, $\operatorname{div} F(p) = \lim_{A \rightarrow 0} A^{-1} \oint_{\partial A} F \cdot n \, ds$, for regions shrinking regularly to p .

7. Divergence in orthogonal coordinates

In three orthogonal coordinates u^1, u^2, u^3 , the metric is diagonal and $\sqrt{g} = h_1 h_2 h_3$. If $F_{(i)}$ are the components in the unit basis $\hat{e}_i = \partial_i / h_i$, then $F_{(i)} = h_i V^i$. Substitution into the coordinate formula gives

Physical components in an orthonormal basis

$$\operatorname{div} F = \frac{1}{h_1 h_2 h_3} \left[\partial_1 (h_2 h_3 F_{(1)}) + \partial_2 (h_1 h_3 F_{(2)}) + \partial_3 (h_1 h_2 F_{(3)}) \right].$$

Polar coordinates in two dimensions

Here $h_r = 1$, $h_\theta = r$, and $\sqrt{g} = r$. For unit-basis components F_r, F_θ ,

$$\operatorname{div} F = \frac{1}{r} \frac{\partial(r F_r)}{\partial r} + \frac{1}{r} \frac{\partial F_\theta}{\partial \theta}.$$

Cylindrical coordinates

With $(h_\rho, h_\phi, h_z) = (1, \rho, 1)$ and $\sqrt{g} = \rho$,

$$\operatorname{div} F = \frac{1}{\rho} \frac{\partial(\rho F_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z}.$$

Spherical coordinates

With $(h_r, h_\theta, h_\phi) = (1, r, r \sin \theta)$ and $\sqrt{g} = r^2 \sin \theta$,

$$\operatorname{div} F = \frac{1}{r^2} \frac{\partial(r^2 F_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sin \theta F_\theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi}.$$

8. The Laplacian and the role of the Hodge star

For a scalar function, the **Laplace–Beltrami operator** is the divergence of the gradient:

$$\Delta f = \operatorname{div}(\operatorname{grad} f).$$

In Cartesian Euclidean coordinates this becomes the familiar sum $\Delta f = f_{xx} + f_{yy} + f_{zz}$. A different construction is needed in the language of forms because the exterior derivative is nilpotent:

$$df = (\partial_i f) du^i, \quad d(df) = d^2 f = 0.$$

Applying d twice therefore cannot give the Laplacian. The metric enters through the Hodge star between the two differentiations.

Intrinsic construction on functions

On an oriented Riemannian n -manifold,

$$\Delta f = \star d \star df.$$

The degrees in the construction are

$$\Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{\star} \Omega^{n-1} \xrightarrow{d} \Omega^n \xrightarrow{\star} \Omega^0.$$

The first d measures change of f ; $\star df$ represents the associated flux form; the next d measures its net outward flux; the final star returns a scalar.

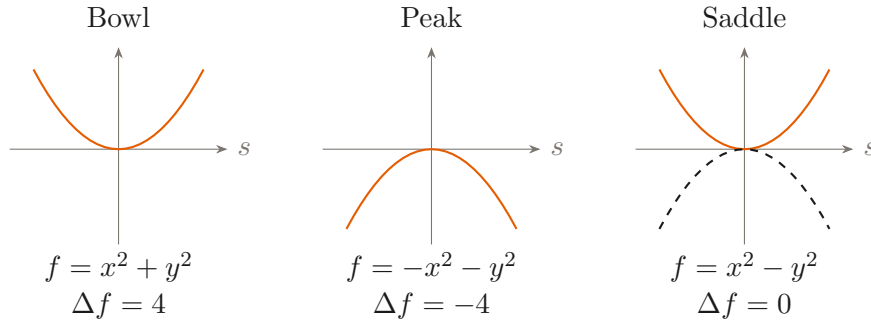


Figure 4: The Laplacian adds directional curvatures. These are vertical slices through the origin (common vertical display scale): solid orange is $f(s, 0)$; the dashed curve is $f(0, s)$ when it differs. For a saddle, positive and negative curvatures cancel, so $\Delta f = 0$ does not mean f is constant. In two Euclidean dimensions, the mean over a radius- ε circle satisfies $\bar{f}_\varepsilon - f(p) = \frac{\varepsilon^2}{4} \Delta f(p) + o(\varepsilon^2)$.

Rotational invariance

Euclidean rotations preserve the metric. Since the exterior derivative is coordinate independent and the Hodge star is determined by the metric and orientation, the Laplacian commutes with rotations R :

$$\Delta(f \circ R) = (\Delta f) \circ R.$$

Consequently, rotating a solution of $\Delta f = 0$ produces another solution. This is the rotational symmetry of Laplace's equation.

9. Laplace–Beltrami in coordinates

The gradient has components $(\text{grad } f)^i = g^{ij} \partial_j f$. Inserting these into the divergence formula gives, for $g = \det(g_{ij}) > 0$,

General curvilinear coordinates

$$\Delta f = \frac{1}{\sqrt{g}} \partial_i (\sqrt{g} g^{ij} \partial_j f).$$

In orthogonal coordinates, $g_{ii} = h_i^2$ and $\sqrt{g} = \prod_j h_j$, so

$$\Delta f = \frac{1}{\prod_j h_j} \sum_i \partial_i \left(\frac{\prod_j h_j}{h_i^2} \partial_i f \right).$$

Polar coordinates (r, θ)

With $(h_r, h_\theta) = (1, r)$,

$$\Delta f = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \theta^2}.$$

Cylindrical coordinates (ρ, ϕ, z)

With $(h_\rho, h_\phi, h_z) = (1, \rho, 1)$,

$$\Delta f = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}.$$

Spherical coordinates (r, θ, ϕ)

With $(h_r, h_\theta, h_\phi) = (1, r, r \sin \theta)$,

$$\begin{aligned} \Delta f = & \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) \\ & + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}. \end{aligned}$$

10. Curl and circulation

In three dimensions, the **curl** measures local circulation per unit area and points along the oriented axis of that circulation. In a fixed Cartesian orthonormal basis, for $F = (F_x, F_y, F_z)$,

$$\begin{aligned}\nabla \times F &= (\partial_y F_z - \partial_z F_y) \hat{e}_x + (\partial_z F_x - \partial_x F_z) \hat{e}_y \\ &\quad + (\partial_x F_y - \partial_y F_x) \hat{e}_z.\end{aligned}$$

The symbolic determinant with rows $(\hat{e}_x, \hat{e}_y, \hat{e}_z)$, $(\partial_x, \partial_y, \partial_z)$, and (F_x, F_y, F_z) packages this Cartesian expression. In curvilinear coordinates the basis itself varies, so its Cartesian form cannot be reused without scale factors.

Definition through differential forms

For an oriented Riemannian 3-manifold, the construction lowers the vector index, differentiates the resulting 1-form, takes its Hodge dual, and raises the index again:

Intrinsic curl in three dimensions

$$F \xrightarrow{\flat} F^\flat \in \Omega^1 \xrightarrow{d} dF^\flat \in \Omega^2 \xrightarrow{\star} \star dF^\flat \in \Omega^1 \xrightarrow{\sharp} \text{curl } F.$$

In short,

$$\boxed{\text{curl } F = (\star dF^\flat)^\sharp.}$$

If $F^\flat = (F^\flat)_i du^i$, its exterior derivative is

$$dF^\flat = \sum_{i < j} [\partial_i (F^\flat)_j - \partial_j (F^\flat)_i] du^i \wedge du^j.$$

This 2-form records oriented circulation through infinitesimal area elements.

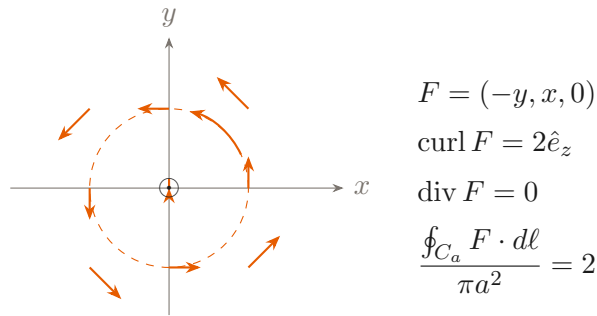


Figure 5: Circulation without a source. This field rotates counterclockwise; the dot denotes the positive z direction, out of the page. On a radius- a circle, the tangential speed is a , giving circulation $2\pi a^2$ and circulation per area 2. The field is tangent to the circle, so its outward flux is zero. Curl and divergence detect different local behavior.

11. Curl in orthogonal coordinates

Let (u^1, u^2, u^3) be positively oriented orthogonal coordinates with scale factors h_i and physical components $F_{(i)}$ in the unit basis $\hat{e}_i$. The familiar determinant mnemonic becomes

$$\nabla \times F = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_1 & h_2 \hat{e}_2 & h_3 \hat{e}_3 \\ \partial_1 & \partial_2 & \partial_3 \\ h_1 F_{(1)} & h_2 F_{(2)} & h_3 F_{(3)} \end{vmatrix}.$$

Expanding it shows each component and its associated scale factors:

$$\begin{aligned} (\nabla \times F)_{(1)} &= \frac{\partial_2(h_3 F_{(3)}) - \partial_3(h_2 F_{(2)})}{h_2 h_3}, \\ (\nabla \times F)_{(2)} &= \frac{\partial_3(h_1 F_{(1)}) - \partial_1(h_3 F_{(3)})}{h_1 h_3}, \\ (\nabla \times F)_{(3)} &= \frac{\partial_1(h_2 F_{(2)}) - \partial_2(h_1 F_{(1)})}{h_1 h_2}. \end{aligned}$$

Polar plane (r, θ)

For a planar field with unit-basis components F_r, F_θ , the oriented curl is normal to the plane:

$$\nabla \times F = \frac{1}{r} \left[\frac{\partial(r F_\theta)}{\partial r} - \frac{\partial F_r}{\partial \theta} \right] \hat{e}_z.$$

Equivalently, the coefficient of $\hat{e}_z$ is a scalar curl in two dimensions.

Cylindrical coordinates (ρ, ϕ, z)

With $(h_\rho, h_\phi, h_z) = (1, \rho, 1)$,

$$\begin{aligned} \nabla \times F &= \left(\frac{1}{\rho} \partial_\phi F_z - \partial_z F_\phi \right) \hat{e}_\rho + (\partial_z F_\rho - \partial_\rho F_z) \hat{e}_\phi \\ &\quad + \frac{1}{\rho} [\partial_\rho(\rho F_\phi) - \partial_\phi F_\rho] \hat{e}_z. \end{aligned}$$

Spherical coordinates (r, θ, ϕ)

With $(h_r, h_\theta, h_\phi) = (1, r, r \sin \theta)$,

$$\begin{aligned} \nabla \times F &= \frac{\partial_\theta(\sin \theta F_\phi) - \partial_\phi F_\theta}{r \sin \theta} \hat{e}_r \\ &\quad + \frac{\frac{1}{\sin \theta} \partial_\phi F_r - \partial_r(r F_\phi)}{r} \hat{e}_\theta \\ &\quad + \frac{\partial_r(r F_\theta) - \partial_\theta F_r}{r} \hat{e}_\phi. \end{aligned}$$

12. Fundamental identities from $d^2 = 0$

The exterior derivative satisfies $d^2\omega = 0$ for every smooth form. This single property explains two familiar vector-calculus identities on an oriented Riemannian 3-manifold.

The curl of a gradient vanishes

Because $(\text{grad } f)^\flat = df$,

$$\text{curl}(\text{grad } f) = (\star d(\text{grad } f)^\flat)^\sharp = (\star d^2 f)^\sharp = 0.$$

A gradient field therefore has no infinitesimal circulation.

The divergence of a curl vanishes

Since $(\text{curl } F)^\flat = \star dF^\flat$ and $\star^2 = 1$ on 2-forms in Riemannian dimension three,

$$\text{div}(\text{curl } F) = \star d \star (\text{curl } F)^\flat = \star d \star \star dF^\flat = \star d^2 F^\flat = 0.$$

Thus a curl field has zero local source density.

13. Operator map

These four constructions share d , the metric-dependent musical maps, and the Hodge star. Their domains and geometric meanings are distinct.

Operator	Input $\rightarrow$ output	Geometric meaning
Gradient	Scalar $\rightarrow$ vector	Direction and rate of greatest increase, measured by the metric.
Divergence	Vector $\rightarrow$ scalar	Net outward flux per unit volume.
Curl in 3D	Vector $\rightarrow$ vector	Oriented local circulation per unit area.
Laplacian	Scalar $\rightarrow$ scalar	Divergence of the gradient; local second-order variation.

Intrinsic expressions and degree changes

$$\begin{aligned}
 \text{grad } f &= (df)^\sharp, & \Omega^0 &\xrightarrow{d} \Omega^1 \xrightarrow{\sharp} \mathfrak{X}(M), \\
 \text{div } F &= \star d \star F^\flat, & \mathfrak{X}(M) &\xrightarrow{\flat} \Omega^1 \xrightarrow{\star} \Omega^{n-1} \xrightarrow{d} \Omega^n \xrightarrow{\star} \Omega^0, \\
 \text{curl } F &= (\star dF^\flat)^\sharp \quad (n=3), & \mathfrak{X}(M) &\xrightarrow{\flat} \Omega^1 \xrightarrow{d} \Omega^2 \xrightarrow{\star} \Omega^1 \xrightarrow{\sharp} \mathfrak{X}(M), \\
 \Delta f &= \star d \star df, & \Omega^0 &\xrightarrow{d} \Omega^1 \xrightarrow{\star} \Omega^{n-1} \xrightarrow{d} \Omega^n \xrightarrow{\star} \Omega^0.
 \end{aligned}$$

Here $\mathfrak{X}(M)$ denotes vector fields. The divergence and Laplacian formulas use the Riemannian sign convention $\Delta = \text{div grad}$.